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Questions

Q1.

The first cyclotron was created by E O Lawrence in 1931.

Lawrence obtained an expression for the time t for a particle of charge q and mass m to complete a circular path in a uniform magnetic field of magnetic flux density B.

The expression obtained by Lawrence is

$$t = \frac{2\pi m}{Bq}$$

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(a) (i) Derive this expression.

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(ii) In Lawrence's cyclotron, an alternating potential difference was used to accelerate particles to high energies.

Explain the significance of the expression obtained by Lawrence to the operation of a cyclotron.

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(b) In a cyclotron, in which Lawrence's expression applies, protons were accelerated to a speed of 1.5×10^7 m s^{-1}.

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Determine the time taken to accelerate a proton to this speed from rest.
magnetic flux density = 1.6 T
accelerating potential difference = 13 kV

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Time = ...

(c) Explain why high energy particles are required to investigate the structure of nucleons.

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(Total for question = 15 marks)

Q2.

The photograph shows a hospital radiotherapy machine that contains a linear accelerator (linac). The linac accelerates electrons to very high speeds for use in treating cancer.

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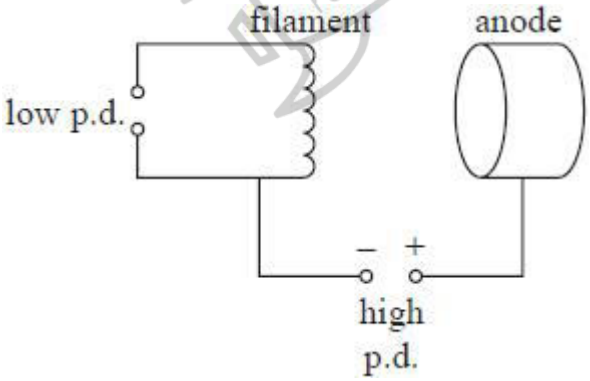
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(© Alexander Tihonov/Pearson Asset Library)

(a) The electrons used in the linac are produced by the filament in an electron gun, as shown in the diagram.



Name and describe the process in which electrons are produced by the filament.

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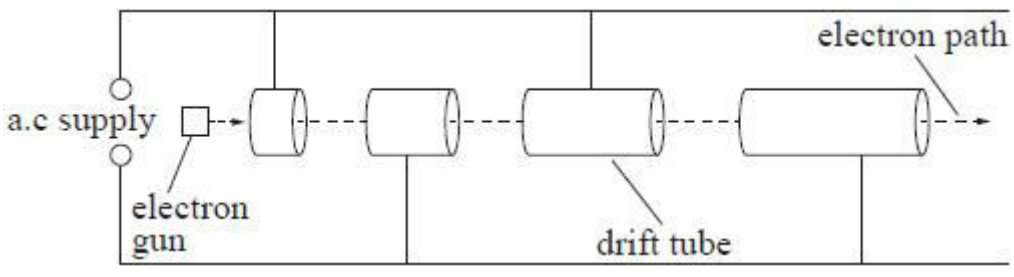
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(b) The diagram shows part of a simplified linac.



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An electron enters the linac from the electron gun with a speed of 2.5×10^6 m s^{-1}. The electron passes through 60 drift tubes before emerging from the other end.

(i) Calculate the energy of the electron as it emerges.
potential difference between adjacent drift tubes = 80 kV

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Energy of the electron = ...

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(ii) For very high energy electrons, successive drift tubes have the same length.
Explain why.

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(c) Linacs used for radiotherapy operate by means of a standing wave formed within the linac.

Explain how standing waves are produced.

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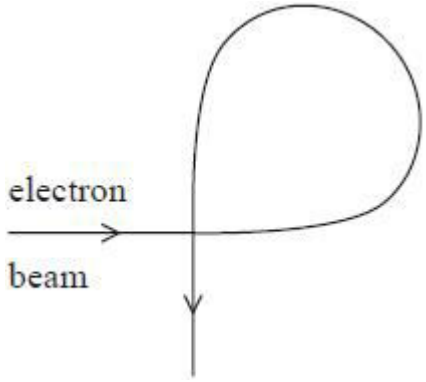
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(d) Electrons emerging from a standing wave linac must be directed towards the patient. Magnets are used to rotate the path of the electron beam through 270°.

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Determine the magnetic flux density required to rotate the beam through 270°.
energy of electrons = 2.5 keV
radius of curvature of path = 61 mm

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Magnetic flux density = ...

(Total for question = 14 marks)

Q3.

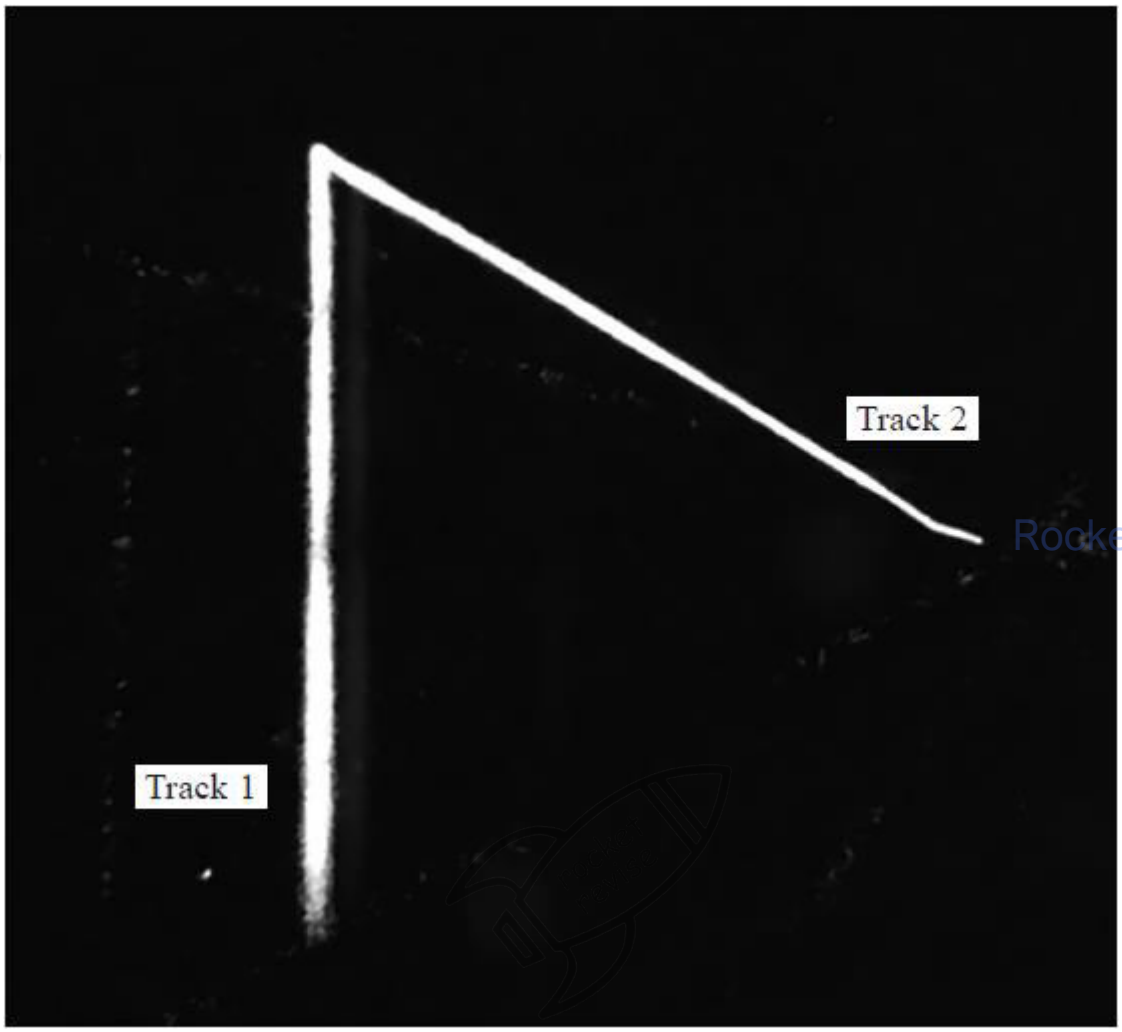
Cloud chambers are used to observe the paths of particles.

The photograph shows a pair of tracks made by alpha particles emitted when an atom of radon decays in a cloud chamber.

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(a) The radon atoms do not leave tracks.

State a reason for this.

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(b) Radon decays by emitting an alpha particle, producing a polonium nucleus. The polonium then decays almost immediately by emitting another alpha particle, producing a lead nucleus.

(i) Complete the nuclear equation for this sequence of decays.

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$$^{220}_{\ldots\ldots}\mathrm{Rn} \rightarrow {}^{\ldots\ldots}_{82}\mathrm{Pb} + {}^{\ldots\ldots}_{\ldots\ldots}\alpha$$

(ii) The radon atom was stationary before the first alpha emission. The angle between the two alpha tracks is 60°.

Sketch a vector diagram to show the momentum of the two alpha particles and the lead ion.

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(iii) Determine the magnitude of the velocity of the lead ion.
velocity of alpha particle emitted by radon = 1.74×10^7 m s^{-1}
velocity of alpha particle emitted by polonium = 1.81×10^7 m s^{-1}
mass of lead ion = 3.52×10^{-25} kg

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Magnitude of velocity = ...

(c) The photograph shows another pair of tracks following radon decay.



Deduce which track is for the first alpha particle emitted.

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(Total for question = 14 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structure of nucleons can be investigated using electrons with high energies.

Which of the following is the reason why high energies are required?

- ☒ **A** to allow for the creation of new particles
- ☒ **B** to overcome the repulsive electrostatic forces
- ☒ **C** to produce a relativistic increase in particle lifetime
- ☒ **D** to produce short de Broglie wavelengths

(Total for question = 1 mark)

Q5.

A positron and a muon travel at the same speed through a magnetic field.
The magnetic field is directed into the page.

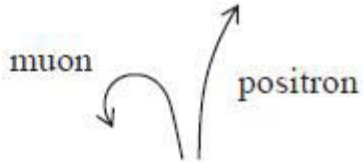
Which of the following diagrams best represents the motion of the particles?

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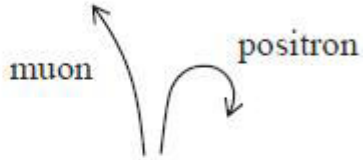
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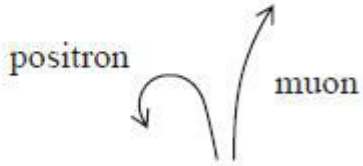
☒ **A**



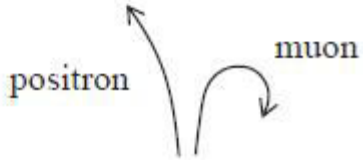
☒ **B**



☒ **C**



☒ **D**



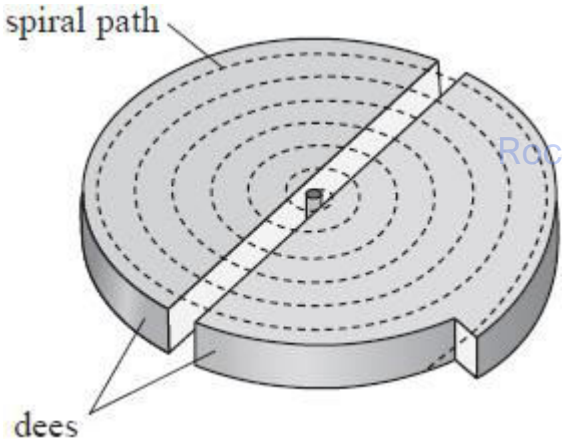
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(Total for question = 1 mark)

Q6.

Select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A charged particle is accelerated in a spiral path in a cyclotron.



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Which of the following changes would increase the amount of energy transferred to the particle as it travels through 360° ?

☒ **A** increasing the distance between the dees
☒ **B** increasing the frequency of the potential difference across the dees
☒ **C** increasing the strength of the magnetic field within the dees

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☒ **D** increasing the maximum potential difference across the dees

(Total for question = 1 mark)

Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A linac is a particle accelerator. It consists of drift tubes of increasing length with potential differences (p.d.) applied between them.

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Which of the following is the reason the drift tubes are of increasing length?

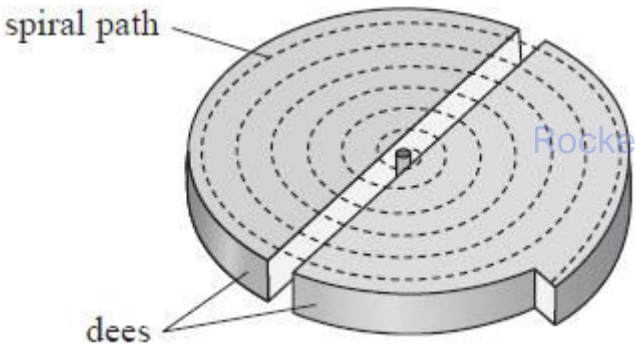
☒ **A** So that the p.d. can have a constant frequency.
☒ **B** So that the particle gains the same amount of energy in each tube.
☒ **C** So that the p.d. between the tubes increases for each successive tube.
☒ **D** So that the time spent in each successive tube increases.

(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In a cyclotron, a charged particle is accelerated in a spiral path.



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Which of the following increases as the radius of the spiral path increases?

☒ **A** The frequency of the alternating p.d. across the dees.
☒ **B** The magnetic flux density acting perpendicularly to the spiral path.
☒ **C** The potential difference between the dees.
☒ **D** The velocity of the charged particles.

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(Total for question = 1 mark)

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Electrons can be used to probe the nuclei of atoms.

Which of the following best explains this?

- ☒ **A** Electrons can be accelerated to very high energies.
- ☒ **B** Electrons can be deflected by a magnetic field.
- ☒ **C** Electrons can be part of an atom.
- ☒ **D** Electrons can be released by thermionic emission.

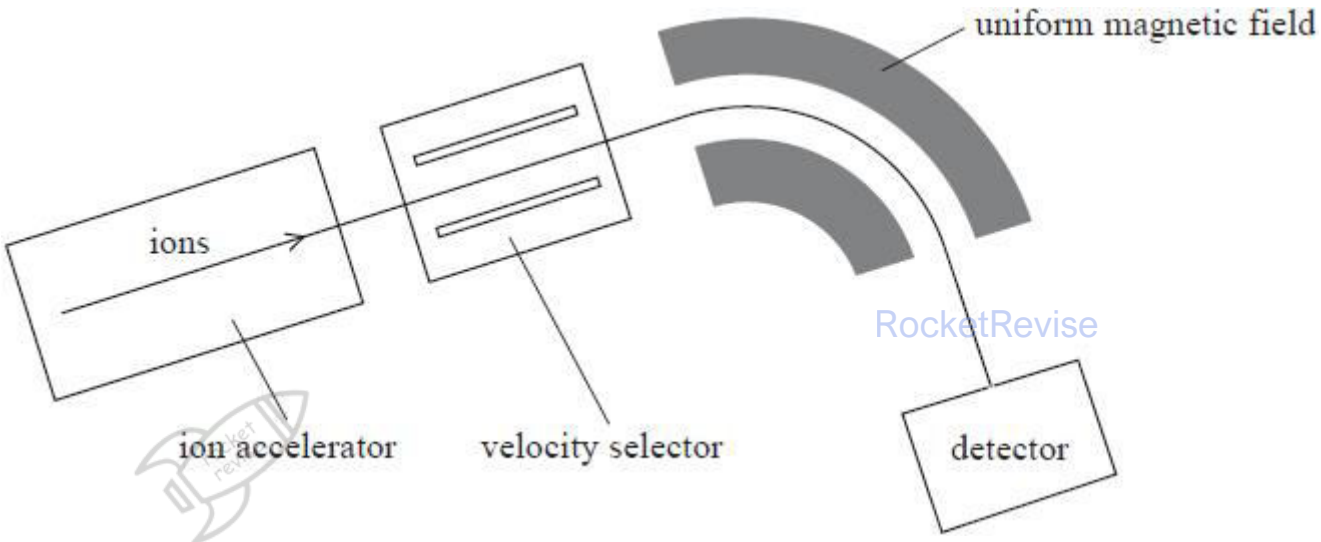
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(Total for question = 1 mark)

Q10.

Mass spectrometry is used to determine the masses of different isotopes of an element.

Atoms of the isotopes are ionised and accelerated. They pass through a velocity selector and into a region with a uniform magnetic field, as shown.

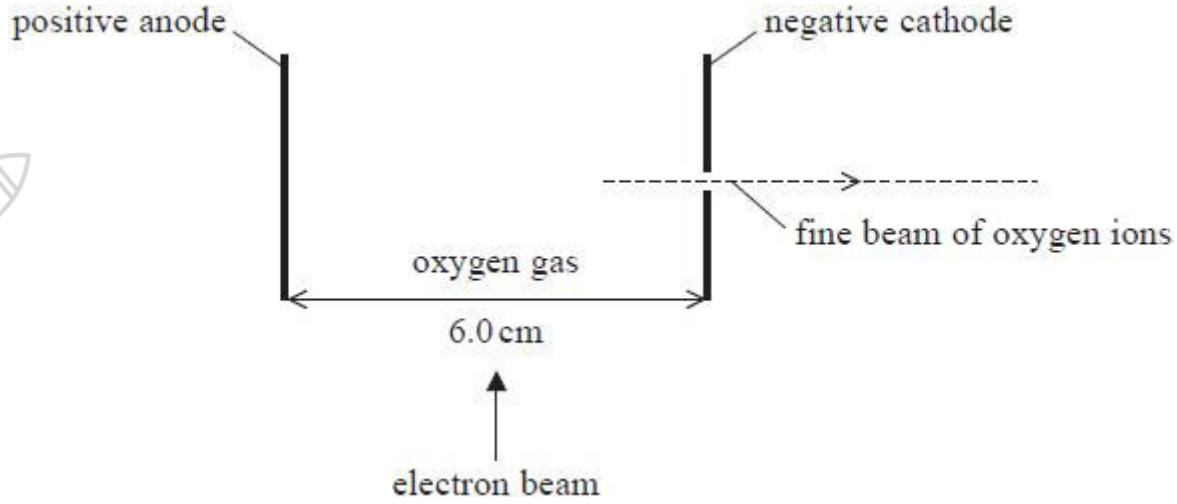


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(a) An ion accelerator uses an electron beam to ionise atoms of oxygen gas. The positive ions are then accelerated across a high potential difference between an anode and a cathode, as shown in the diagram below. The cathode has a hole in it so that the accelerated ions may pass in a fine beam to the velocity selector.

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(i) The electric field strength between the anode and cathode is 7.5×10^5 V m^{-1}.
Calculate the potential difference between the anode and cathode.

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Potential difference = ...

(ii) A stationary oxygen ion is formed half-way between the anode and cathode with a charge of $+1.6 \times 10^{-19}$ C.
Show that the velocity of the ion as it passes through the hole in the cathode is about 5×10^5 m s^{-1}.
mass of oxygen ion = 2.7×10^{-26} kg

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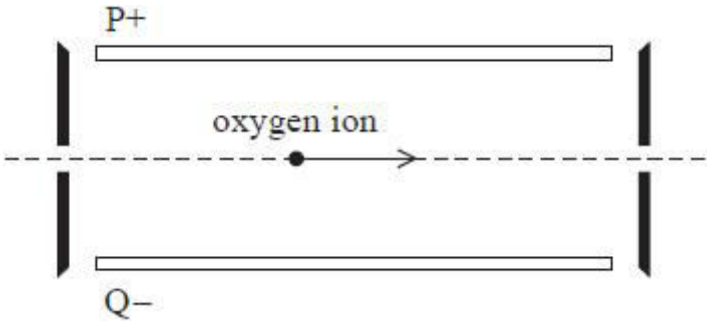
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(b) Oxygen ions enter the velocity selector with a range of velocities. The velocity selector allows ions with a specific velocity to travel in a straight line, as shown. Plate P is positive and plate Q is negative.



(i) A uniform electric field between the plates acts on an oxygen ion. A uniform magnetic field acts so

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that the magnetic force on the oxygen ion is in the opposite direction to the electric force.
Explain the direction of the magnetic field.

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(ii) The velocity selector is used to produce a beam of oxygen ions travelling in a straight line with a speed of 5.0×10^5 m s^{-1}.
Calculate the magnetic flux density of the magnetic field that is required.
electric field strength between plates = 10 500 N C^{-1}

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Magnetic flux density = ...

(c) After passing through the velocity selector, a beam of oxygen ions with the same velocity enters a region of uniform magnetic flux density. Different isotopes of oxygen can be present in the beam.

Explain why the detector will only detect one particular isotope.

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(Total for question = 14 marks)

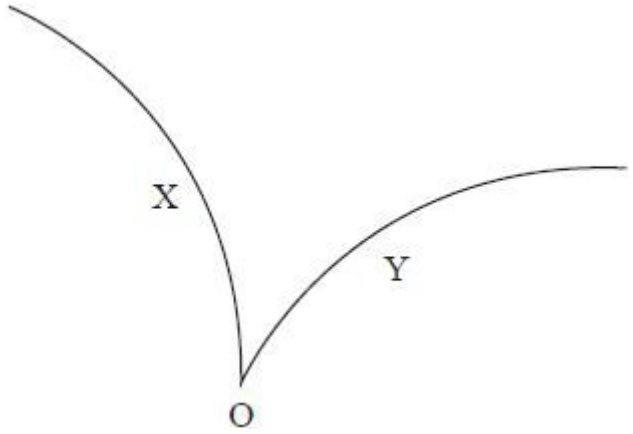
Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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A particle detector shows tracks produced by two particles X and Y. The particles were created by the decay of a kaon at O.



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Which of the following can be concluded from the tracks?

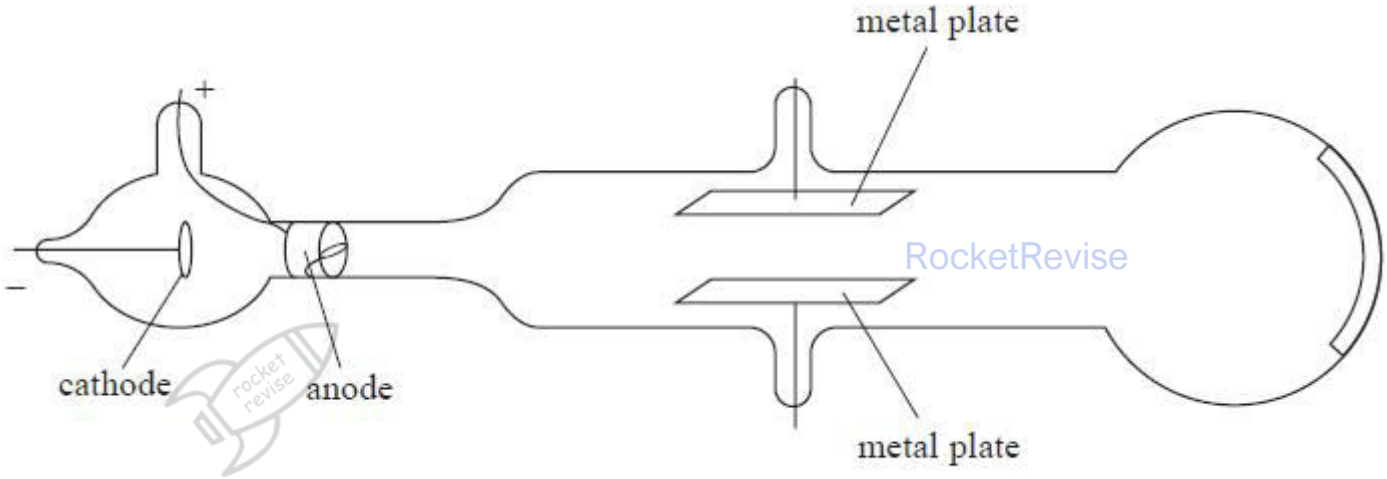
- ☒ **A** A magnetic field acts into the page.
- ☒ **B** X is a positively charged particle.
- ☒ **C** Y is an electron.
- ☒ **D** X and Y have opposite charge.

(Total for question = 1 mark)

Q12.

(a) In 1897 J J Thomson demonstrated that electrons are small negative particles.

The diagram shows the apparatus used by Thomson.



(Source: J.J. Thomson © Philosophical Magazine, 44, 293 (1897))

A potential difference *V* was applied across the metal plates, producing an electric field between them.
A magnetic field of flux density *B* was applied at right angles to the electric field.
Electrons were emitted from the cathode and accelerated towards the anode. The electrons moved between the metal plates with a speed *v*.
Thomson adjusted *B* until the electric and magnetic forces on the electrons were equal and opposite, so the electrons passed between the metal plates in a straight line.

(i) The plates are separated by a distance *d*.

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Show that the speed v of the electrons is given by

$$v = \frac{V}{Bd}$$

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(ii) An experiment was carried out using similar apparatus.
Show that v was about 3×10^7 m s^{-1} .
d = 1.5 cm
$B = 5.5 \times 10^{-4}$ T
V = 231 V

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(iii) When the electric field between the plates is switched off the electrons move in a circular path of radius 39 cm, due to the magnetic field.
The accepted value for the charge per unit mass of an electron is 1.8×10^{11} C kg^{-1}.
Deduce whether the charge per unit mass of an electron calculated using data from this experiment is consistent with the accepted value.
$B = 5.5 \times 10^{-4}$ T

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(b) In 1927 J J Thomson's son directed beams of electrons at thin films of metal.

The photograph shows one of the patterns observed.

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Explain how this pattern changed scientists' understanding about the nature of electrons.

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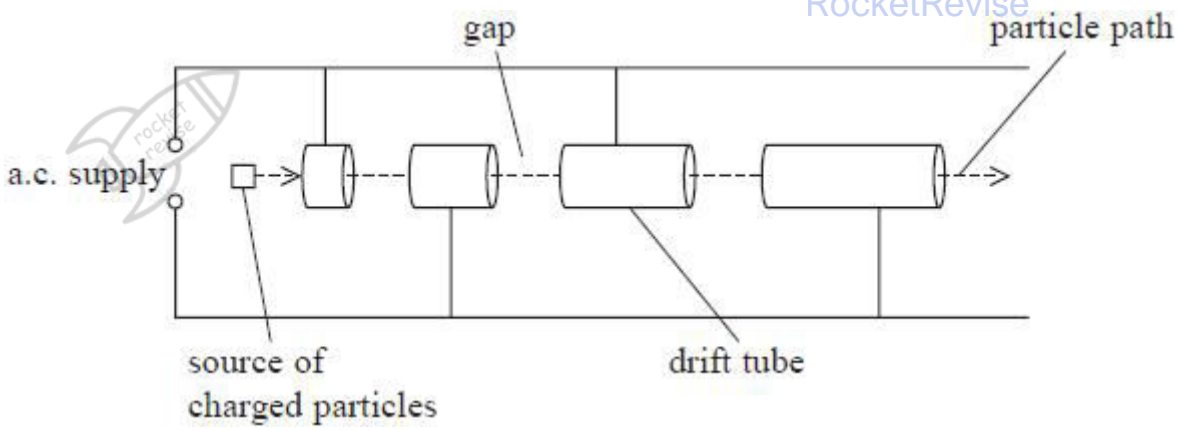
(Total for question = 11 marks)

Q13.

Linacs and cyclotrons both accelerate charged particles to very high speeds.

(a) The diagram shows a linac.

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Explain the use of electric fields in a linac.
You should refer to the frequency of the a.c. supply.

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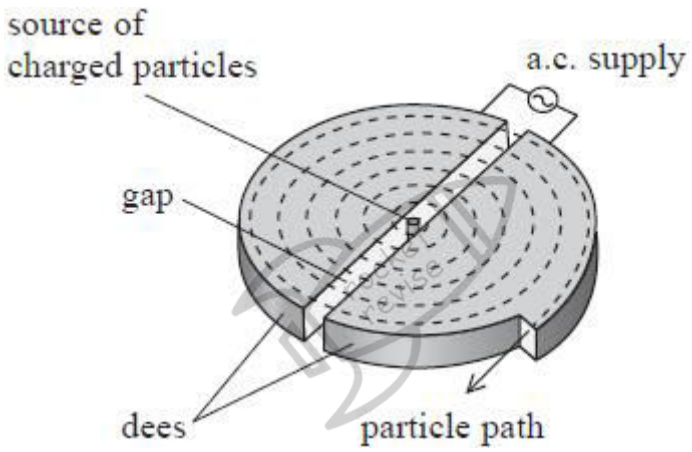
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(b) The diagram shows a cyclotron.

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Explain why a magnetic field is applied at right angles to the dees in the cyclotron.

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(Total for question = 6 marks)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Scientists use particles with high energy to investigate the structure of nucleons.

Which of the following statements is **not** a reason why particles with high energy are required?

☒ **A** to allow forces between particles to be overcome

☒ **B** to ensure particles have a very high momentum

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☒ **C** to ensure particles have a very small de Broglie wavelength
☒ **D** to provide sufficient energy for the production of new particles

(Total for question = 1 mark)

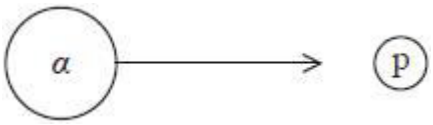
Q15.

In the 1930s, scientists investigated collisions of alpha particles with protons to determine whether the collisions were elastic.

The diagrams show an alpha particle before and after a collision with a stationary proton. The proton moves off at an angle θ to the original direction of the alpha particle.

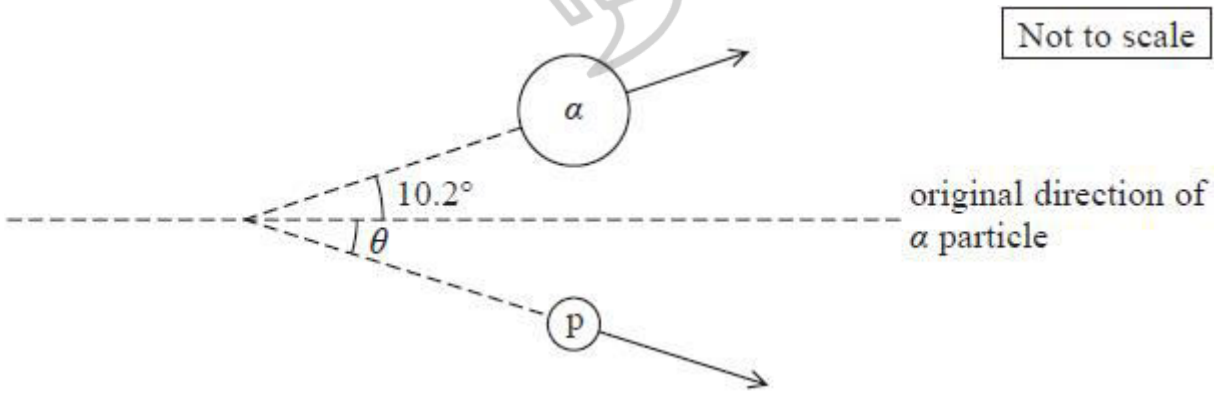
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Before collision



momentum of alpha particle = 1.26×10^{-19} N s

After collision



momentum of alpha particle = 8.06×10^{-20} N s

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(a) Show that the momentum of the proton after the collision is about 5×10^{-20} N s at an angle θ, where θ is about 20°.

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(b) Deduce whether the collision was elastic.

mass of alpha particle = 6.64×10^{-27} kg

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(Total for question = 10 marks)

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Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows the paths of two charged particles, X and Y, moving at the same speed in a uniform magnetic field.

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Which row of the table describes the properties of Y compared with the properties of X?

	Mass of Y	Charge of Y
☒ **A**	greater than	the same as
☒ **B**	less than	greater than
☒ **C**	the same as	less than
☒ **D**	greater than	less than

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(Total for question = 1 mark)

Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The tubes of a linear accelerator (linac) get progressively longer along the length of the linac.

Which of the following is the reason for this?

☒ **A** The frequency of the applied potential difference decreases.
☒ **B** The frequency of the applied potential difference increases.
☒ **C** The particles gain mass as they accelerate.
☒ **D** The particles must spend the same time in each tube.

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(Total for question = 1 mark)

Q18.

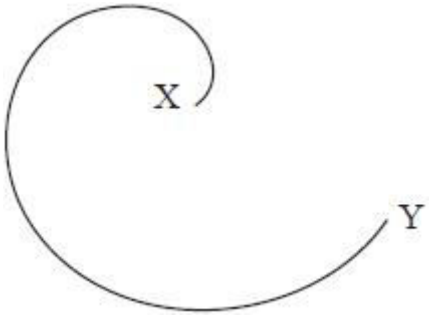
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows the track of a charged particle in a magnetic field.

The magnetic field is directed into the page.

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Which of the following particles could create this track?

- ☒ **A** a muon travelling from X to Y
- ☒ **B** a positron travelling from Y to X
- ☒ **C** a π^+ travelling from X to Y
- ☒ **D** an electron travelling from Y to X

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(Total for question = 1 mark)

Q19.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of the following statements is true for a cyclotron?

- ☒ **A** A cyclotron can be used to accelerate neutrons.
- ☒ **B** A cyclotron uses a magnetic field to increase the speed of particles.
- ☒ **C** A cyclotron uses an alternating potential difference at a constant frequency.
- ☒ **D** A cyclotron uses an electric field to keep particles moving in a circular path.

(Total for question = 1 mark)

Q20.

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Auroras occur above the Earth's poles when charged particles collide with gases in the upper atmosphere.

These collisions produce tiny flashes that fill the sky with light, as shown.

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(Source: © Jamen Percy/Shutterstock)

(a) Collisions between electrons and oxygen atoms cause the emission of green light.

Calculate the velocity of electrons that would cause green light of wavelength 560 nm to be emitted from
an oxygen atom.

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Velocity of electrons = ...

(b) Radio waves are also produced during an aurora.

Electrons moving in a circular path in the atmosphere emit radio waves.
The frequency of the emitted radio waves is the same as the frequency of rotation of the electron.
An electron with a speed of 1650 km s^{-1} is moving perpendicular to a magnetic field with magnetic flux density 48 μT.
A radio station is broadcasting at a wavelength of 220 m.
Deduce whether the radiation emitted by the electron would interfere with the signal from the radio station.
You should calculate the radius of the circular path of the electron.

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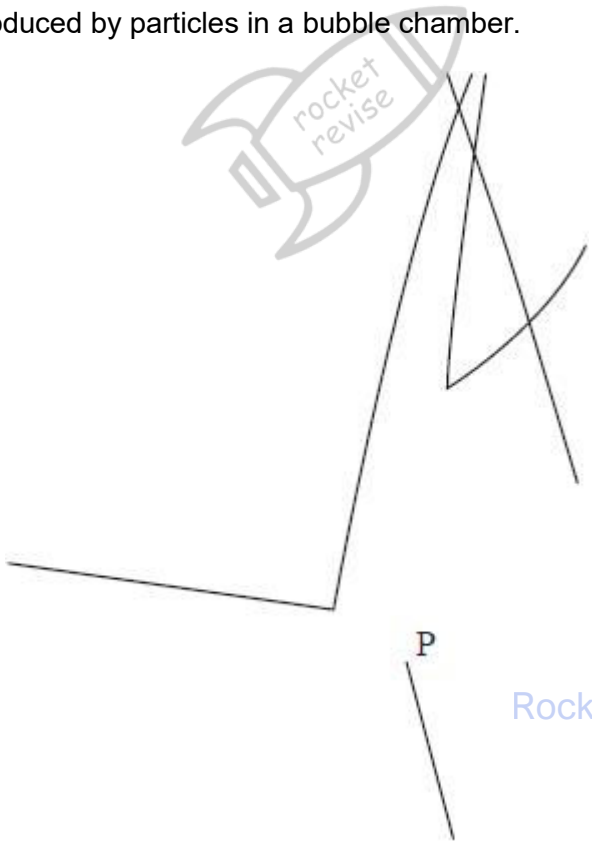
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(Total for question = 8 marks)

Q21.

The diagram shows tracks produced by particles in a bubble chamber.



A charged particle interacts with a stationary proton at P. This produces two neutral particles. Each of these neutral particles is unstable and decays into other particles.

Explain the evidence provided by the diagram for this description of the particle interactions.

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(Total for question = 3 marks)

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Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structure of nucleons can be investigated using high energy electrons.

Which of the following is the reason high energies are needed?

- ☐ **A** to cause annihilation
- ☐ **B** to produce individual quarks
- ☐ **C** to produce short de Broglie wavelengths
- ☐ **D** to overcome the repulsive electrostatic forces

(Total for question = 1 mark)

Q23.

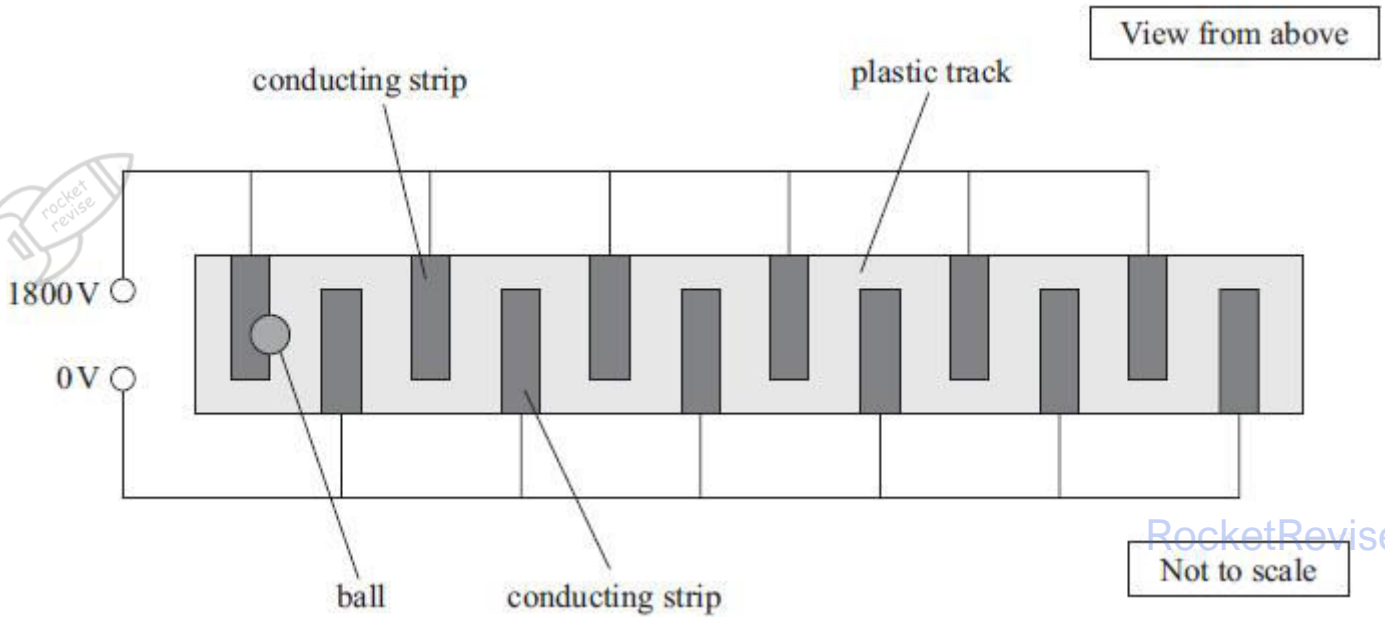
The diagram shows some equipment used by a teacher to demonstrate the action of electric fields.

A long white plastic track has strips of conducting material attached to it. The strips on one side of the track are connected to a potential of 1800 V. The strips on the other side are connected to a potential of 0 V.

A ball with a conducting surface can be accelerated along the track.

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(a) The teacher places the ball on the first strip. The teacher switches on the power supply and the ball accelerates.

The size of the gap between the strips is 2.0 cm.

(i) The ball accelerates uniformly across the gap between the first strip and the second strip in 0.22 s.
Show that the acceleration of the ball is about 0.8 m s^{-2}.

(2)

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(ii) When placed on the first strip, the ball is charged to a potential of 1800 V.
Show that the charge on the ball is about 2×10^{-9} C.
radius of ball = 0.95 cm

(2)

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(iii) There is a force on the ball due to the uniform electric field between the strips.
Deduce whether this is the only force acting to accelerate the ball at about 0.8 m s^{-2}.
mass of ball = 0.18 g

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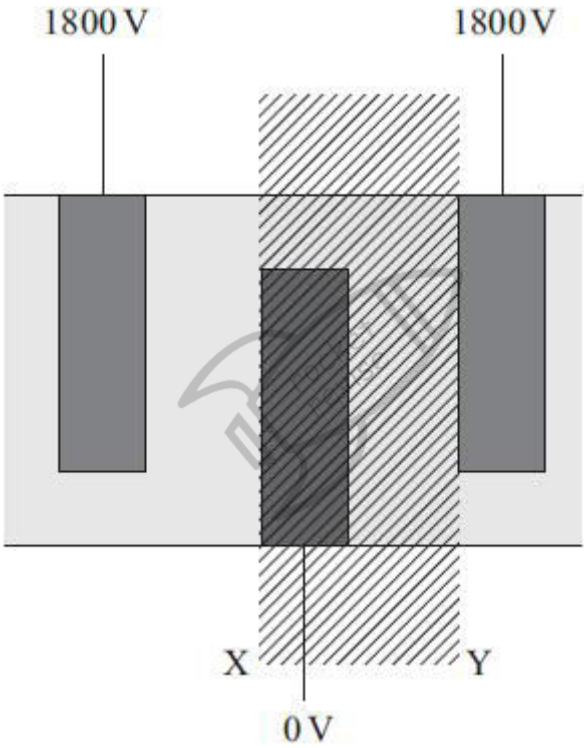
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(b) The equipment can be used as a model to demonstrate the acceleration of charged particles in a linac.

When placed on the first strip, the ball is charged to a potential of 1800 V. The ball then accelerates towards the next strip at 0 V, where it discharges.
The ball then moves at constant speed to the next 1800 V strip. The ball is charged again and accelerates to the next 0 V strip.
This pattern of movement continues until the ball reaches the end of the track.
The diagram shows part of the model.



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(i) The shaded section of the diagram between X and Y corresponds to the drift tube in a linac.
Give **one** similarity and **two** differences in the way this section of the model works and the way a drift tube in a linac works.

(3)

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(ii) In the model, the lengths corresponding to drift tubes are constant.
Explain why the lengths of the drift tubes in a linac are not constant.

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(Total for question = 14 marks)

Q24.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Cyclotrons accelerate charged particles to high speeds. Cyclotrons use electric fields and magnetic fields.

Which of the following statements is **not** correct?

- ☒ **A** The magnetic field is at right angles to the velocity of the particles.
- ☒ **B** The magnetic fields in the dees keep the particles following circular paths.
- ☒ **C** The particles are accelerated by electric fields between the dees.
- ☒ **D** The particles are accelerated by electric fields inside the dees.

(Total for question = 1 mark)

Q25.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

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A proton moving in a straight line with a speed v enters a magnetic field. The magnetic field is perpendicular to the direction of motion of the proton.

The proton follows a circular path of radius r in the magnetic field.

The path of the proton then decreases in radius.

Which of the following is a reason why the radius decreases?

- ☒ **A** The kinetic energy of the particle decreases.
- ☒ **B** The magnetic flux density decreases.
- ☒ **C** The mass of the proton increases.
- ☒ **D** The momentum of the proton increases.

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(Total for question = 1 mark)

Q26.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

An alpha particle also moving in a straight line with the same speed *v* enters the same magnetic field.

The alpha particle follows a circular path in the magnetic field.

Which of the following is the initial radius of the path?

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- ☐ **A** $4r$
- ☐ **B** $2r$
- ☐ **C** $\frac{r}{2}$
- ☐ **D** $\frac{r}{4}$

(Total for question = 1 mark)

Q27.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Cyclotrons use electric fields and magnetic fields to give particles very high energy.

Which of the following statements is true for charged particles in cyclotrons?

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- ☐ **A** The particles are accelerated by electric fields and magnetic fields.
- ☐ **B** The particles follow circular paths of constant radius.
- ☐ **C** The electric field changes direction when the particles are in the gap between the dees.
- ☐ **D** The magnetic field changes direction when the particles are in the gap between the dees.

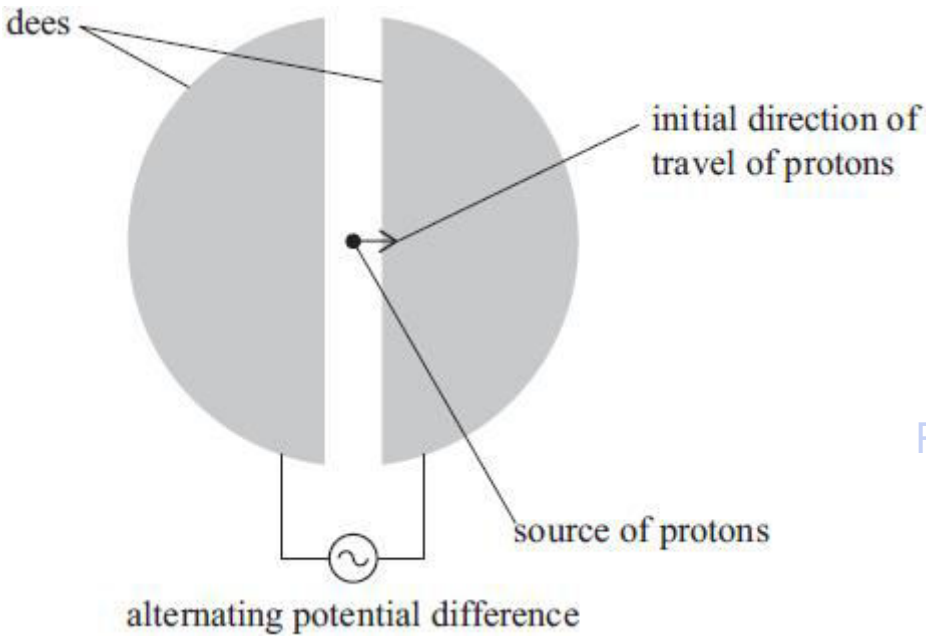
(Total for question = 1 mark)

Q28.

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A simplified diagram of a cyclotron is shown. In the cyclotron, protons gain energy.
Protons are produced in the source and initially travel in the direction shown.



A magnetic field is applied perpendicular to the dees. The magnetic field direction is into the page.

(a) Add to the diagram to show the path of the protons as they gain energy in the cyclotron.

(2)

(b) The frequency f of the alternating potential difference applied to the dees is given by the equation

$$f = \frac{Bq}{2\pi m}$$

where
m is the mass of the proton
B is the magnetic flux density
q is the charge on the proton.
Calculate the time a proton spends in one of the dees.
B = 0.55T

(3)

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Time = ...

*(c) Explain how the cyclotron increases the energy of the protons.

You should refer to the alternating potential difference and the magnetic field.

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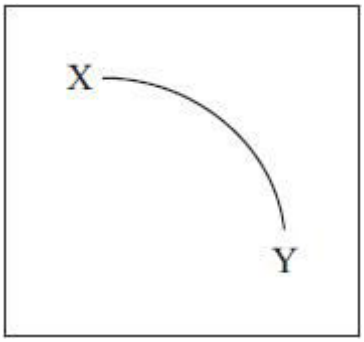
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(Total for question = 11 marks)

Q29.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows a particle track in a bubble chamber.



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Which of the following is a valid conclusion from this track?

- ☒ **A** The particle is negatively charged.
- ☒ **B** The particle is moving from X to Y.
- ☒ **C** The speed of the particle is increasing.
- ☒ **D** The particle experiences a centripetal force.

(Total for question = 1 mark)

Q30.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

In high-energy collisions, electrons are used to investigate the structure of nucleons.

Which of the following is the reason why the electrons need a very high energy?

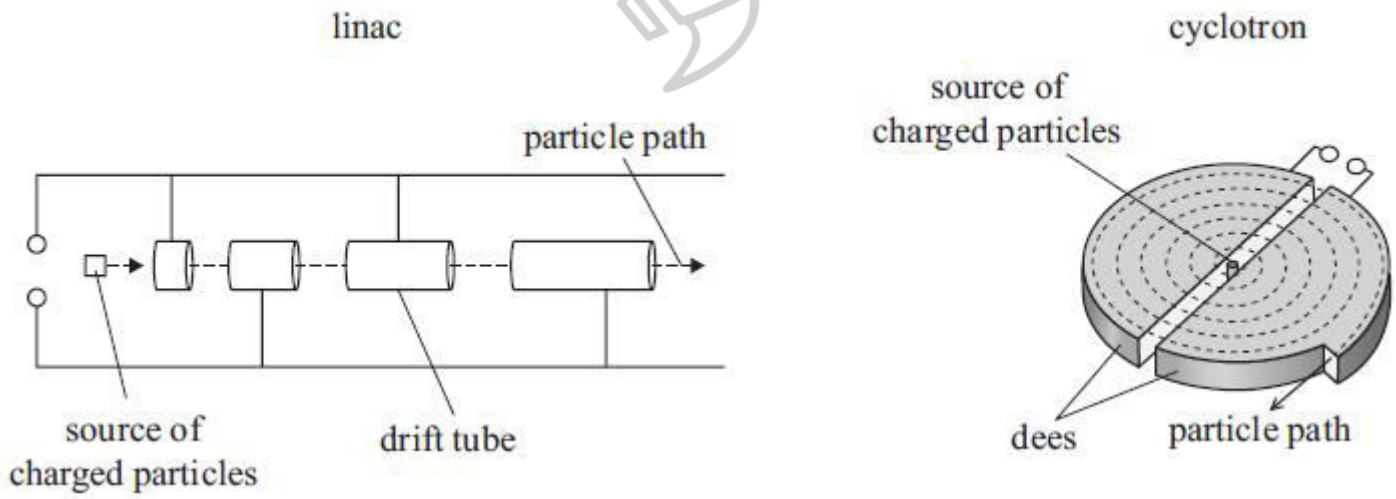
- ☒ **A** So that new particles are created.
- ☒ **B** So that collisions between electrons and nucleons are elastic.
- ☒ **C** So that the electrons have a large wavelength.
- ☒ **D** So that a quark is knocked out of the nucleon.

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(Total for question = 1 mark)

Q31.

The simplified diagrams show a linac and a cyclotron. Both machines are used to accelerate charged particles.



*(a) Explain how these machines use a potential difference to accelerate charged particles.

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(b) A scientist plans to use each type of particle accelerator to accelerate protons from rest.

(i) In the linac, a potential difference of 0.40 MV is applied to the tubes. The protons pass through 4 gaps.
Show that the speed of the protons is about 2×10^7 m s^{-1} as they leave the 4th gap.

(4)

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(ii) In the cyclotron, the protons leave the dees when the radius of their path is 29 cm.
Deduce whether the speed of the protons as they leave the cyclotron is the same as the speed of the protons as they leave the 4th gap in the linac.
magnetic flux density = 0.55 T

(3)

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(Total for question = 13 marks)

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